

Association Between Breathlessness Severity and Quality of Life Among Autorickshaw Drivers in Jalgaon: A Cross-Sectional Study

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Abstract:

➤ Background:

Chronic respiratory symptoms, particularly breathlessness, can have broader effects on an individual's daily functioning and well-being, extending beyond physical limitations to psychological, social, and environmental aspects of quality of life. Autorickshaw drivers may be at increased risk of respiratory symptoms due to prolonged exposure to traffic-related pollutants and other occupational conditions, which may subsequently be associated with poorer quality of life.

➤ Objective:

To determine the association between breathlessness severity and quality of life among autorickshaw drivers in Jalgaon.

➤ Methods:

A cross-sectional study involving 104 autorickshaw drivers from Jalgaon was carried out. The WHOQOL-BREF was used to assess quality of life, while the ATS dyspnea grading was used to measure the degree of dyspnea. The quality-of-life scores were summarized using descriptive statistics, and the association between the four WHOQOL-BREF domains and the ATS breathlessness grade was determined using Spearman's correlation. The level of significance for statistical significance was $p < 0.05$.

➤ Results:

The social interactions domain had the highest mean score (54.50 ± 8.98) while the environmental domain had the lowest mean score (41.81 ± 8.14). ATS breathlessness grade showed significant negative correlations with the physical health domain ($\rho = -0.798$, $p < 0.001$), psychological health ($\rho = -0.393$, $p < 0.001$), social relationships ($\rho = -0.353$, $p < 0.001$), and environmental domain ($\rho = -0.420$, $p < 0.001$). The strongest correlation was observed between breathlessness severity and physical health.

➤ Conclusion:

Higher breathlessness severity showed a significant association with reduced WHOQOL-BREF domain scores among Jalgaon autorickshaw drivers, with the strongest relationship observed for physical health. These findings suggest that increasing breathlessness severity is related to changes in multiple dimensions of quality of life within this occupational group.

Keywords: Chronic Breathlessness; Autorickshaw Drivers; Quality of life; WHOQOL-BREF; ATS Dyspnea Grading.

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I. INTRODUCTION

Chronic respiratory problems are long-term conditions that affect the airways and other structures of the lungs, leading to difficulties in breathing and reduced lung function.^[1]

Respiratory symptoms are one of the main reasons for consultations at primary health care units. These include symptoms such as bleeding from the airways, Epistaxis, hemoptysis, Coughing, Breathing, abnormalities, Dyspnea, Wheezing, Excessive breathing, sneezing, Chest and throat discomfort.^[2] Dyspnea is a common respiratory symptom; the Burden of Obstructive Lung Disease (BOLD) study reported dyspnea in approximately 27% of participants across 15 countries.^[3]

Professional drivers are vulnerable to a variety of environmental, occupational, and health-related variables that could affect their quality of life. According to Sanchez et al., taxi and motorcycle taxi drivers experience occupational conditions involving physical and mental demands, prolonged working hours and other health-related challenges.^[5] Since drivers are highly exposed to urban air pollution due to prolonged roadside work, it leads to increased rates of chronic respiratory symptoms including wheezing, persistent coughing, and dyspnea.^[6] A study conducted by Adhikari and Gupta among autorickshaw drivers in Mangalore reported that 28.7% experienced breathlessness, 19.7% had chronic cough, and 32.4% had low peak expiratory flow rate (PEFR) values. a conference proceedings.^[4] Another study by Peter Stephen et al. (2018) among 297 autorickshaw drivers in urban Puducherry also reported a high prevalence of respiratory symptoms, with 68% of drivers experiencing breathlessness on exertion, highlighting the substantial occurrence of breathlessness in this occupational population.^[1] A study examining changes in quality of life with the duration of chronic breathlessness reported that greater breathlessness severity was associated with reduced mobility, limitations in daily activities, and increased pain or discomfort.^[9]

Quality of life (QOL) refers to an individual's perception of their position in life in relation to their cultural context, personal values, goals, expectations, standards, and concerns. Assessing Quality of Life involves looking at multiple aspects including health, psychological factors, environmental considerations, and social issues.^[6] The WHOQOL-BREF is a 26-item instrument developed by the World Health Organization to assess quality of life. It comprises four domains: physical health, psychological well-being, social relationships, and environmental health. Responses are recorded on a five-point Likert scale, with higher scores indicating better perceived quality of life. The domain scores are transformed to a scale ranging from 20 to 100.^[7]

Although respiratory symptoms among autorickshaw drivers have been studied in relation to respiratory health, limited research has examined the association between breathlessness severity and quality of life in this occupational group, particularly in Jalgaon. Therefore, the present study

aimed to determine the association between breathlessness severity and quality of life among autorickshaw drivers in Jalgaon.

II. MATERIALS AND METHODOLOGY

➤ Study Design

An observational, cross-sectional study was conducted among autorickshaw drivers.

➤ Study Setting

The study was conducted among autorickshaw drivers at selected autorickshaw stands in Jalgaon city.

➤ Sample Duration

The study was carried out over a six-month period.

➤ Sampling Method

A convenience sampling method was used for participant selection. A total of 104 autorickshaw drivers participated in the study.

➤ Inclusion and Exclusion Criteria

• Inclusion Criteria

- ✓ Drivers of open-cabin three-wheelers engaged in passenger transportation for a minimum duration of three years.
- ✓ 20-65 years of age autorickshaw drivers.
- ✓ Persons experiencing breathlessness for ≥ 3 years, with a dyspnea grade greater than 0 according to the ATS dyspnea scale.

• Exclusion Criteria

- ✓ Workers with physical disabilities.
- ✓ Anyone who was not willing to participate.
- ✓ Those with any variables that may affect the QOL Those with any variables that may affect the QOL such as past/current respiratory illness, lung surgery etc.

➤ Data Collection Procedure

Participants meeting the eligibility criteria were screened and recruited for the study. The study objectives and procedures were explained to each participant, following which written informed consent was obtained. Demographic details were recorded using a structured data collection form. Breathlessness was then assessed for its presence and severity using the American Thoracic Society (ATS) dyspnea scale. Quality of life was subsequently evaluated using the WHOQOL-BREF questionnaire, comprising four domains: physical health, psychological health, social relationships, and environment. The questionnaire was administered to the participants in a standardized manner, and adequate clarification was provided whenever required without influencing their responses. All collected information was recorded systematically and checked for completeness. The data were subsequently coded, entered, and subjected to appropriate statistical analysis to assess the relationship

between breathlessness severity and the different domains of Quality of life.

III. RESULTS

Statistical analysis was performed using PSPP. Mean and standard deviation were calculated for descriptive analysis. The Shapiro–Wilk test was used to evaluate data distribution and indicated non-normality. Accordingly, non-parametric methods were applied. Spearman’s rank correlation coefficient was used to examine the relationship between breathlessness severity and the four WHOQOL-BREF domains. Statistical significance was set at $p < 0.05$.

➤ Age Wise Distribution

The participants had an age range of 25–64 years, with a mean age of 45.00 ± 8.40 years. The average participant falls directly in mid-adulthood. The standard deviation indicates that roughly 68% of the sample falls between 36.6 and 53.4 years old. (Table 1; Figure 1)

Table 1 Age wise distribution

Variable	Minimum	Maximum	Mean	SD
Age	25.00	64.00	45.00	8.40

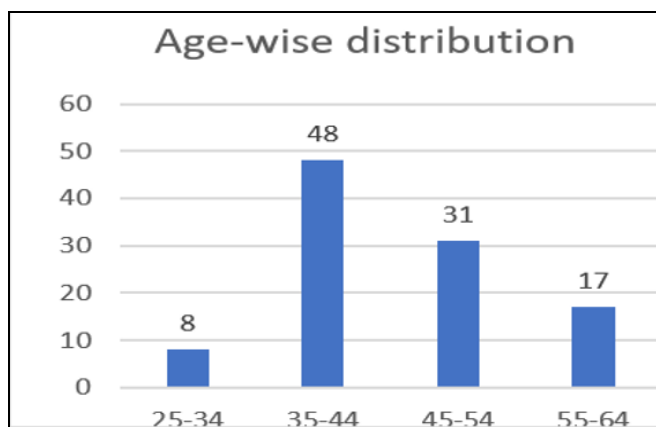


Fig 1 Age Wise Distribution

➤ Quality of Life Domains

• Physical Health:

Yielded a mean score of 43.85, SD = 7.56, range: 31.00 - 63.00, placing it lower than both social and psychological dimensions. The comparatively lower physical health score indicates that physical aspects of quality of life were relatively more affected in the study population.

• Psychological Health:

Scores for psychological well-being averaged 50.38 (SD = 6.81, range: 44.00 - 81.00). The minimum score of 44.00 was the highest baseline floor across all domains, while the maximum score of 81.00 represents the peak score observed in the entire dataset. The lower standard deviation highlights relatively consistent mental health scores across participants.

• Social Relationships:

Participants reported the highest average score among all domains (M = 54.50, SD = 8.98, range: 44.00 - 75.00),

suggesting that interpersonal connections and social support are a relative area of strength for this group. However, it also exhibited the largest standard deviation, indicating the greatest variability in individual experiences.

• Environment:

The environmental domain recorded the lowest mean score overall M = 41.81, SD = 8.14, range: 25.00 - 63.00. It also registered the lowest minimum score 25.00, indicating that aspects such as physical safety, financial resources, living conditions, or transport accessibility may represent a major area of concern or dissatisfaction for the sample.

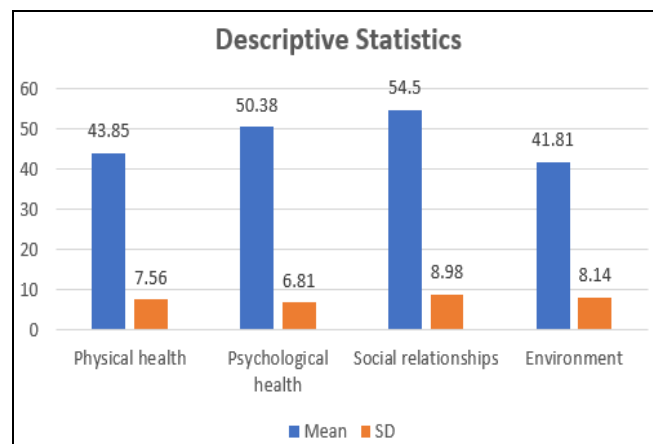


Fig 2 Descriptive Statistics of QOL Domain Score

➤ Ats Grade Distribution

ATS Grade distribution shows that Grade 1 (slight breathlessness) was most common, accounting for 45.2% of the sample (47 participants), followed by Grade 2 (32.7%, 34 participants) and Grade 3 (22.1%, 23 participants). No participants in this sample were classified as Grade 4 (very severe breathlessness). (table 2, figure 3)

Table 2 ATS Grade Distribution

ATS Grade	Frequency	Percent
Grade 1	47	45.2%
Grade 2	34	32.7%
Grade 3	23	22.1%
Grade 4	0	0.0%
Total	104	100.0%

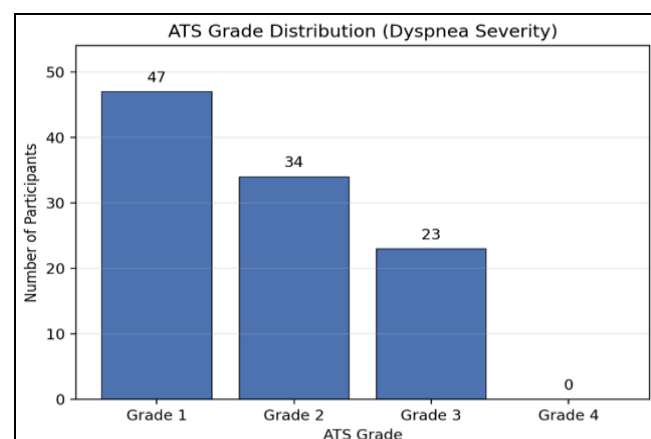


Fig 3 ATS Grade Distribution

➤ Correlation Between QOL Domains and ATS Grading

- A statistically significant negative monotonic association was observed between ATS dyspnoea grade and the WHOQOL-BREF physical health domain score ($\rho = -0.798$, 95% CI $[-0.856, -0.719]$, Holm-adjusted $p < 0.001$). The strength of the association was very high, indicating that participants with greater breathlessness severity tended to have lower physical health scores. Thus, increasing ATS grades were associated with a corresponding reduction in physical health-related quality of life, demonstrating a strong inverse relationship between breathlessness severity and the physical health domain. (figure 4)

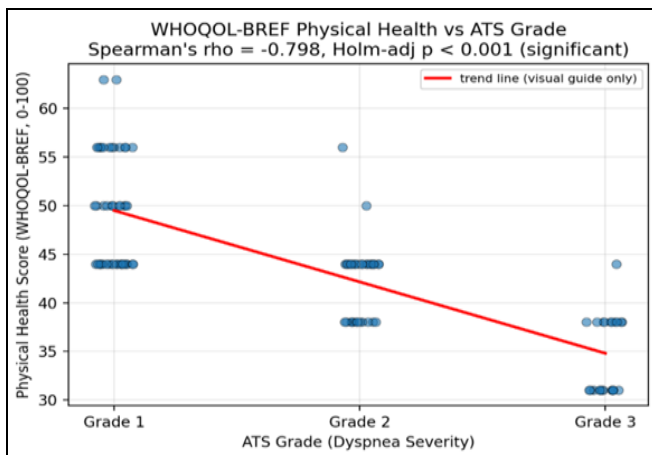


Fig 4 Physical Health vs ATS Grade

- A statistically significant negative monotonic association was identified between ATS dyspnea grade and the WHOQOL-BREF psychological health domain score ($\rho = -0.393$, 95% CI $[-0.557, -0.214]$, Holm-adjusted $p < 0.001$). The correlation was moderate in strength, indicating that higher levels of breathlessness severity were associated with lower psychological health scores. This inverse relationship suggests that increasing breathlessness severity was accompanied by a decline in psychological health-related quality of life among the participants. (figure 5)

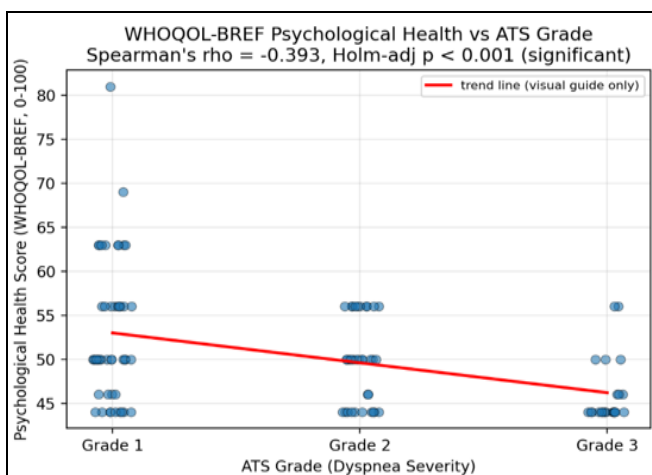


Fig 5 Psychological Health vs ATS Grade

- A statistically significant negative monotonic association was observed between ATS dyspnoea grade and the WHOQOL-BREF social relationships domain score ($\rho = -0.353$, 95% CI $[-0.514, -0.170]$, Holm-adjusted $p < 0.001$). The correlation was moderate in strength, indicating that greater breathlessness severity was associated with lower social relationships scores. This inverse association suggests that participants with higher ATS grades tended to report poorer quality of life in the social relationships domain. (figure 6)

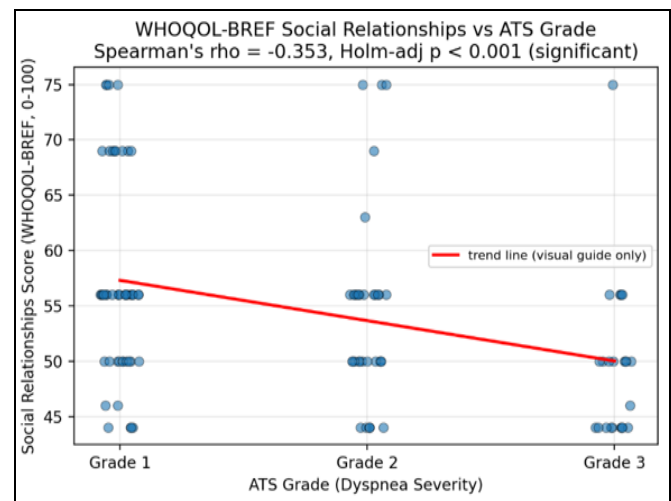


Fig 6 Social Relationship vs ATS Grade

- Be aware of the different meanings of the homophones “affect” and “effect,” “complement” and “compliment,”
- A statistically significant negative monotonic association was found between ATS dyspnea grade and the WHOQOL-BREF environment domain score ($\rho = -0.420$, 95% CI $[-0.569, -0.249]$, Holm-adjusted $p < 0.001$). The moderate strength of this correlation indicates that participants with greater breathlessness severity tended to have lower environment domain scores. Thus, higher ATS grades were associated with poorer quality of life in the environmental domain, demonstrating an inverse relationship between breathlessness severity and environmental quality of life. (Figure 7)

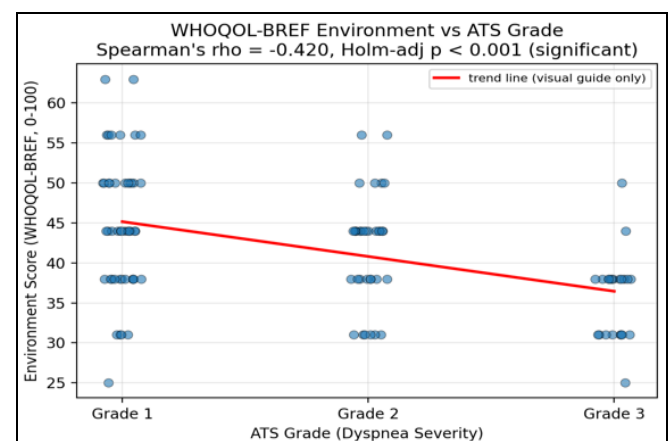


Fig 7 Environmental Domain vs ATS Grade

All four WHOQOL-BREF domains showed statistically significant negative correlations with ATS grade after Holm correction. Physical Health demonstrated the strongest association ($\rho = -0.798$), while Psychological Health, Social Relationships, and Environment showed moderate negative associations ($\rho = -0.35$ to -0.42). Overall, higher breathlessness severity was significantly associated with poorer quality of life across all four domains.

IV. DISCUSSIONS

The present study assessed the quality of life of 104 Jalgaon autorickshaw drivers with chronic respiratory symptoms using the WHOQOL-BREF and examined its association with breathlessness severity using the ATS dyspnea grading. ATS Grade 1 was most common (45.2%), followed by Grade 2 (32.7%) and Grade 3 (22.1%), with no participants classified as Grade 4.

The mean WHOQOL-BREF scores were 43.85 ± 7.56 for physical health, 50.38 ± 6.81 for psychological health, 54.50 ± 8.98 for social relationships, and 41.81 ± 8.14 for environment. The environmental domain had the lowest mean score, while social relationships had the highest. The relatively lower physical health score may be related to the effect of breathlessness on physical functioning and daily activities. This is consistent with Currow et al. (2020), who reported that increasing severity and duration of chronic breathlessness were associated with limitations in mobility and usual activities.^[9] The psychological health findings may also be explained by the effect of persistent respiratory symptoms on daily functioning and well-being. Voll-Aanerud et al. (2010), using data from the European Community Respiratory Health Survey, reported that respiratory symptoms were associated with reduced health-related quality of life, covering both physical and psychological well-being, even among individuals without diagnosed asthma or COPD.^[4]

The social relationships domain showed comparatively higher scores, with approximately equal proportions of participants classified as having good and poor social relationships. This may indicate that many participants maintained social support despite respiratory symptoms. The environmental domain had the lowest mean score, which may be related to occupational factors such as prolonged exposure to traffic-related air pollution, noise, vehicle vibrations and irregular working conditions. This finding is comparable with Sánchez et al., who reported lower environmental quality-of-life scores among taxi drivers.^[5]

A statistically significant inverse correlation was identified between ATS dyspnoea grade and all four WHOQOL-BREF domains. The strongest association was observed with the physical health domain ($\rho = -0.798$, $p < 0.001$), indicating a strong inverse relationship between breathlessness severity and physical quality of life. As the severity of breathlessness increased, physical health scores tended to decrease. This may be explained by the fact that increasing breathlessness can restrict physical activity, mobility, energy, sleep, and the ability to perform activities

of daily living. Moderate negative correlations were observed between ATS grade and psychological health ($\rho = -0.393$, $p < 0.001$), social relationships ($\rho = -0.353$, $p < 0.001$), and environmental quality of life ($\rho = -0.420$, $p < 0.001$). The association with psychological health suggests that greater breathlessness severity may be accompanied by poorer psychological well-being, potentially because persistent breathing difficulty can interfere with daily activities and contribute to emotional distress. Similarly, the negative association with social relationships may indicate that increasing breathlessness can limit participation in social and occupational activities. The association with the environmental domain may reflect difficulties in maintaining normal work and daily living activities in the presence of more severe breathlessness.

The comparatively stronger correlation with the physical health domain, compared with the other three domains, suggests that breathlessness severity is more closely related to the physical aspects of quality of life in this population. However, the significant associations across all four domains indicate that greater breathlessness severity is related not only to physical functioning but also to psychological, social, and environmental dimensions of quality of life. Similar associations between respiratory symptoms, chronic breathlessness and poorer quality of life have been reported by Matheson et al.^[8] These findings support the importance of considering quality of life across multiple domains when evaluating individuals experiencing chronic breathlessness.

Overall, the results indicate that greater breathlessness severity is associated with lower quality of life across the physical, psychological, social, and environmental domains among the autorickshaw drivers included in the study. The observed inverse associations suggest that increasing severity of breathlessness may be accompanied by poorer quality-of-life outcomes across multiple dimensions. Nevertheless, the cross-sectional nature of the study limits the interpretation of these relationships, as temporal sequence and causal relationships between breathlessness severity and quality of life cannot be established. Further research using objective measures of respiratory function, along with longitudinal study designs and follow-up assessments, may help to clarify the progression of these associations and provide a more comprehensive understanding of the relationship between breathlessness severity and quality of life among autorickshaw drivers.

V. CONCLUSIONS

The study concluded that chronic breathlessness was related to poorer quality of life among Jalgaon autorickshaw drivers, as demonstrated by significant negative correlations between ATS breathlessness grade and all four WHOQOL-BREF domains, with the strongest association observed in physical health.

LIMITATIONS

The study was conducted among autorickshaw drivers from Jalgaon, which may restrict the applicability of the findings to autorickshaw drivers from different geographical regions or to individuals engaged in other occupations.

The study assessed quality of life using WHOQOL-BREF but did not include objective pulmonary function measures to quantify respiratory impairment.

Other factors that may influence quality of life, such as occupational exposure, duration of driving, smoking history, and socioeconomic factors, may not have been fully accounted for.

FUTURE SCOPE

Future research may consider recruiting a more diverse sample of autorickshaw drivers from multiple cities and geographical regions to improve the broader applicability of the findings.

Future research can incorporate spirometry and other pulmonary function tests to objectively assess respiratory function.

The relationship between quality of life and duration of occupation, daily driving hours, and years of exposure can be investigated.

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